The MEW nozzle in the round, with heat

What decides whether the melt reaches the collector, for the set-up of Ju, Omar, Mirihanage and Kilic

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Summary. The notes of 3 and 5 November modelled the nozzle as a planar slice, with the melt held at 120 °C. Here the nozzle is round: the model uses its axial symmetry, so the melt's surface has both of its curvatures, which a planar slice lacks and a Taylor cone needs. The melt can also cool. The set-up is otherwise the same: PCL at 820 Pa s, fed at 0.0208 $mm^3\,s^{-1}$ through a 0.4 mm nozzle 5.5 mm above the collector, at 6.5 kV. Four results:

- **Held at** 120 °C, the field draws the drop into a column that reaches the collector after 15.8 s on 50 µm cells and 9.8 s on 25 µm cells. The time is not converged and cannot be: the model's perfectly conducting tip sharpens without limit on finer cells. It is not a prediction, here or in the planar and 3-D runs.
- **Allowed to cool**, with estimated values for the air (25 °C) and for the heat it draws from the surface (90 $W\,m^{-2}\,K^{-1}$), the drop's lower end falls below 40 °C after 23 s and starts to crystallise. The drop then stops about 3.5 mm above the collector while melt piles up at the tip, 1.34 mm^3 within 1 mm of it after 150 s. That is oozing, not a jet, and it holds on finer cells.
- **The air at the tip decides between the two.** With the air at 35 °C the drop still stops; at 40 and 50 °C it reaches the collector after 31 and 28 s. Once the drop cools, the field, the surface tension and the shape of the electrodes change little.
- **The printer does make fibres**, so something at its tip must differ from these estimates: most simply, the air there is warmer than 25 °C, or PCL crystallises lower than assumed, or the jet is started in a way the model leaves out. Four questions would settle it (Section 5).

1 Model and set-up

Geometry	The round nozzle, through its axial symmetry: a two-dimensional model in radius and height (r–z) that keeps the curvature around the axis a planar slice loses. A 0.4 mm bore and an assumed 1.0 mm tip face, 5.5 mm above a grounded collector at 6.5 kV. The collector is at rest, since a moving plate is not axially symmetric; nothing after contact is used.
Melt	PCL at 120 °C: 820 Pa s (measured), 1080 $kg\,m^{-3}$, surface tension 0.03 $N\,m^{-1}$ (nominal). The whole drop conducts, as the estimated conductivity implies (note of 5 November), so the planar model's conducting depth is gone.
Feed	0.0208 $mm^3\,s^{-1}$, delivered exactly in volume.
Field	The potential in the air, with the melt at the nozzle's potential. The air is a cylinder 4 mm in radius whose side and top carry no field; a long needle and a plate across the top were run as checks (Section 2.3).
Heat	New. The melt conducts and carries heat (conductivity 0.20 $W\,m^{-1}\,K^{-1}$, heat capacity 2000 $J\,kg^{-1}\,K^{-1}$, both nominal) and loses it through its surface to air at 25 °C, at 90 $W\,m^{-2}\,K^{-1}$ (nominal: natural convection and radiation). Nozzle at 120 °C, collector at 22 °C. The viscosity rises as the melt cools (Arrhenius, 31.3 $kJ\,mol^{-1}$ from the literature: 8 times its 120 °C value at 50 °C, 17 times at 30 °C), and between 40 and 30 °C the melt crystallises, modelled as a drag that brings it to rest.
Grid	50 µm cells, with 25 µm as a check; 57 µs steps. A free-surface lattice Boltzmann model of the melt (the air is not simulated), and finite volumes for the potential and the temperature on the same cells.

Matched to the experiment, as before: the capillary number of the feed (4.5), the Bond number (0.014) and the ratio of electric to capillary pressure (62).

2 Results

2.1 Held at 120 °C: a column, but no converged time

With the melt held at 120 °C, as in the planar and 3-D runs, the field draws the drop into a column (Fig. 1a, blue; Fig. 1b) that reaches the collector after 15.8 s on 50 µm cells and 9.8 s on 25 µm cells. The finer cells resolve a sharper tip: around 9 s the electric pull there is about 42 times the capillary pressure of the bore, against 6 on the coarser cells, and the drop is 5 mm above the collector after 3.2 s rather than 3.8 s. Nothing in the model

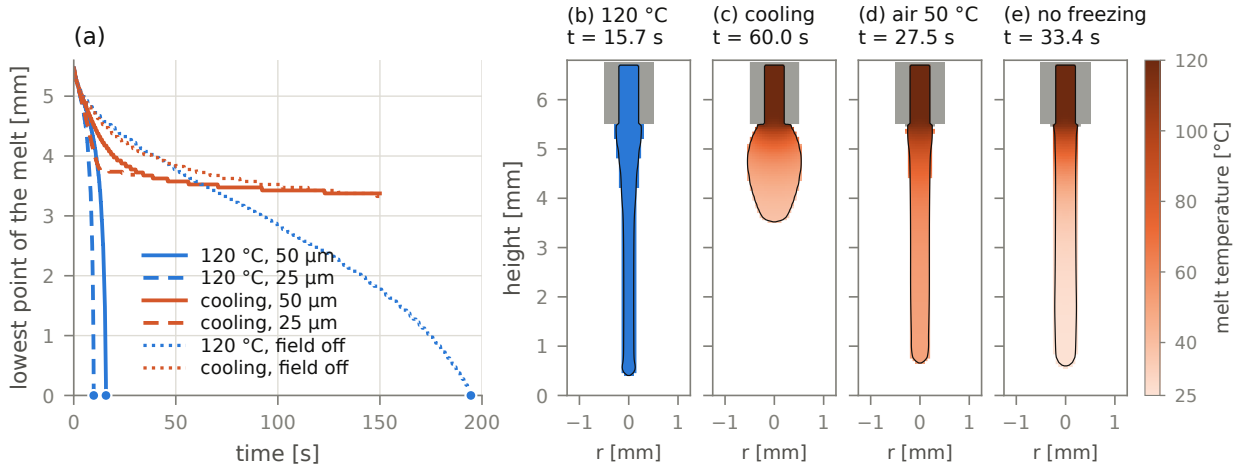


Figure 1. (a) Height of the lowest point of the melt against time, held at 120 °C (blue) and allowed to cool (orange), on 50 µm (solid) and 25 µm cells (dashed), and with the field off (dotted, 50 µm). Dots mark first contact with the collector. (b–e) The drop, mirrored about the axis, coloured by temperature (blue: held at 120 °C), with the nozzle in grey: (b) held at 120 °C, just before contact; (c) cooling, stopped; (d) cooling in air at 50 °C and (e) without the freezing drag, both just before contact. The two cooling runs with the field on in (a) cap the viscosity's rise at 6-fold, the rest at 24-fold; under the two caps the nominal run stays within a cell from 10 to 60 s.

limits a perfect conductor's tip: no current leaves with the melt, and the air does not break down, although the field at the drop, 11 to 43 kV mm^{-1} as it approaches, is far above the roughly 3 kV mm^{-1} at which air breaks down. The contact time is therefore not a prediction. With the field off the drop sags under its own weight, is 1.78 mm above the collector after 150 s, and touches after 195 s. The planar model, with its earlier surface treatment, touched after 91 s: a planar drop holds less melt for its outline.

2.2 Allowed to cool: the drop stops and oozes

With heat (Fig. 1a, orange; Fig. 1c), the melt leaving the bore cools from its surface. Its lower end falls below 45 °C after 18 s and below 40 °C after 23 s, enters the freezing band and stiffens, and the drop all but stops: 3.53 mm above the collector after 60 s, 3.38 mm after 150 s. It widens instead, to 1.64 mm across, and the melt piles up at the tip, 1.34 mm^3 within 1 mm of it after 150 s, while the melt below the tip averages 51 °C. On 25 µm cells, taken to 30 s, the drop is 3.69 mm up after 28 s against 3.78 mm on 50 µm cells, 0.77 mm wide against 0.76 mm, with 6 % less melt at the tip. Unlike the column, this result holds on the finer cells.

With the field off the cooling drop comes down more slowly but stops at the same height (Fig. 2c): 0.25 mm higher after 60 s, at the same 3.43 mm after 120 s, and 3.33 mm up after 150 s with 1.54 mm^3 at the tip. Letting the melt ooze for 60 s and then switching the field on does not make a jet: after 120 s the drop is one cell lower than with the field off (3.38 against 3.43 mm) and holds less at the tip (1.20 against 1.32 mm^3), with a field of about 10 kV mm^{-1} where it is strongest on its surface. Once the drop has cooled, the field barely moves it.

2.3 What decides it

Figure 2 changes the estimates one at a time. The air at the tip decides between the two outcomes (Fig. 2a). At 35 °C the drop still stops, 2.7 mm up; at 40 and 50 °C its end does not reach the freezing band, and it reaches the collector after 30.5 and 28.1 s as a cold column (Fig. 1d). The drop stops when the air can cool its end into the band. Without the freezing drag it also reaches the collector, after 34.3 s (Fig. 1e): the rising viscosity slows the melt but does not stop it. The cooling sets where it stops (Fig. 2b). At 30, 90 and 300 W m^{-2} K^{-1} the drop stops 1.9, 3.5 and 4.3 mm above the collector, holding 0.47, 0.75 and 1.17 mm^3 within 1 mm of the tip after 60 s: the stronger the cooling, the sooner the end freezes and the more melt is held at the tip. A freezing band of 20 to 30 °C in place of 30 to 40 °C lets the drop down to 2.6 mm. Surface tension 20 % higher or lower leaves the drop in the same cell and the melt at the tip within 0.001 mm^3.

The electrodes matter less. Held at 120 °C, in a box 8 mm in radius, a long needle (10 mm into the box) touches after 15.1 s, and a plate at the nozzle's potential across the top, 5 mm up the nozzle, after 18.2 s, against 15.8 s in the default box: a pull at the tip 35 % weaker delays contact by 15 to 20 %.

3 What the model leaves out

The jet: the preprint's fibres are 9.5 to 22 µm across, below the finest cells that can be afforded (25 µm), so the model resolves what gathers under the tip and nothing that leaves it. **Charge and current**: the melt is a perfect conductor, no current leaves with it, and the air does not break down. **The moving collector**: at rest, so nothing after contact is used. **Crystallisation** is a drag switched on below a temperature band, with no

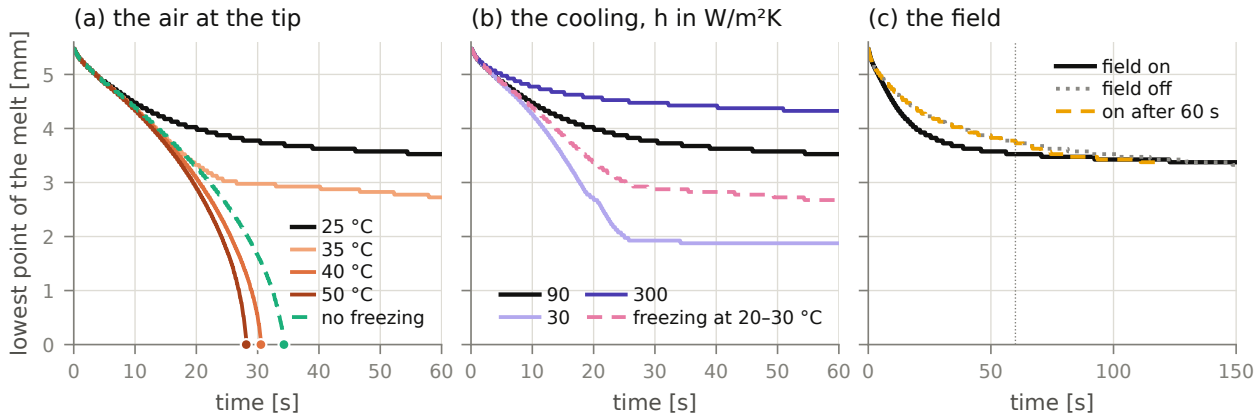


Figure 2. What decides it, on 50 µm cells: the lowest point of the melt against time, changing one estimate at a time. Black is the nominal run (air at 25 °C, 90 W m^{-2} K^{-1}, freezing at 30 to 40 °C); dots mark first contact. (a) The air at the tip, and the run without the freezing drag. (b) The heat drawn from the surface, and a lower freezing band. (c) The field off, and switched on after 60 s (dotted line). The field-on run in (c) caps the viscosity's rise at 6-fold, the others at 24-fold.

kinetics and no latent heat; the heat released as the melt crystallises would slow its end through the band. **The heat values**: the heat drawn from the surface, the air's temperature, the conductivity and heat capacity, the freezing band and the activation energy are all estimates.

4 How far to trust it

Against the 3-D run of 6 November (held at 120 °C, its own settings)	In a cylinder inscribed in its square box the column reaches 0.5 mm above the collector after 31.1 s, against 32.4 s in 3-D, and the melt between tip and collector agrees to 0.09 % over 1 to 32 s; in a cylinder of the same area, 27.2 s, the box's shape moving the field at the tip. 7.6 min of computing against 6.0 h.
Cell size, 50 and 25 µm	Cooling: 3.69 mm against 3.78 mm after 28 s. Held at 120 °C: contact after 9.8 and 15.8 s, not converged.
The model's own viscosity setting	Contact (120 °C) after 15.45, 15.69 and 15.80 s as it goes from 1/2 to 1/4 to 1/6, against 16.07 s extrapolated: about 2 %.
Flow, field and heat against exact solutions	Laplace's law for a sphere, 1.0 % high at 8 cells and 0.03 % at 16; pipe flow, second order; a drop relaxing to a sphere at 0.98 to 0.99 of the exact rate. A conducting sphere's field to 3 % in amplitude, with a 7 % cell-to-cell scatter from its stepped surface; coaxial electrodes, second order. Conduction in a cylinder to 0.01 % at 32 cells; a sphere's heat loss 1.0 % to 1.2 % high.
Surface stiffness	Under a gentle electric pull the surface now deflects 4 to 6 % too far (planar test); before the improved curvature used here it deflected 9 to 20 % too little.
Mass	The fed melt is accounted for to 3×10^{-5} after the first 2 s (1.4×10^{-5} on 25 µm cells); to 2.3×10^{-4} at the viscosity setting 1/2, and to 1.2×10^{-3} in the comparison with 3-D, which uses that setting and the 3-D run's others.

5 Questions for the experiment

1. What is the air temperature a few millimetres below the tip while printing? A thermocouple would do. The model's answer changes between 35 and 40 °C.
2. After the voltage is switched on, does a jet start by itself, and after how long? If it needs help, how is it started?
3. Is there a DSC trace of this PCL cooling at a few kelvin per second (the model's drop end averages about 3.4 K s^{-1} from 120 to 40 °C)? Its crystallisation onset sets the band, 30 to 40 °C here, and its latent heat is missing from the model.
4. Could the viscosity data at 90, 100 and 110 °C be shared? They would replace the activation energy taken from the literature.

M3LB (github.com/alexderosis/M3LB, branch `electrowriting`): `demonstrator/mew_axi` with `-heat`, adding `-nu 0.0416666667` for the 24-fold viscosity cap (the default, 1/6, caps at 6); `-Tair 35, 40, 50`; `-nofreeze`; `-h 30, 300`; `-Tfreeze 20`; `-gamma 0.024, 0.036`; `-nofield`; `-fieldon 60`. Electrodes `-W 8 -up 10` and `-W 8 -up 5 -plateR 8`; 25 µm cells `-R 8`; against 3-D `-W 2` (or 2.26) `-equip -lc 5 -nu 0.5 -nogeom -feed 0.02159`. Series and logs in `results/mew_axi`.